

Werner Bisschoff

ENTERPRISE SYSTEMS & AUTOMATION ARCHITECT

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Strategic Enterprise Software Architect with 5+ years of experience engineering robust business automation platforms, custom ERP extensions, and scalable API layers. Expert in the Frappe/ERPNext ecosystem, asynchronous workflow systems, and cross-platform mobile application deployment (Expo/React Native). Proven track record of translating complex real-world operations into scalable, type-safe data schemas governed by strict Spec-Driven Development (SDD) paradigms.

Experience

Principal Systems Architect

Cape Town

DIVERGENT TABLETOP

Jul 2025

- Engineered and hosted an enterprise-grade documentation ecosystem.
- Implemented a single-database multi-tenant separation engine using PostgreSQL Row-Level Security (RLS) and transaction-scoped local parameters (SET LOCAL), enforcing strict data silos across isolated groups.
- Designed an atomic asynchronous flush strategy using in-memory GenServer storage buffers, reducing database persistent write transactions by 85% during periods of high user interaction.
- Utilized Ecto.Multi atomic chains and Oban transactional background workers to ensure multi-stage backend edits (e.g., structural document renaming and incoming backlink indexing) execute completely or roll back safely.
- Applied Universal Design and Executive Functioning support models to product UX/UI; established highly structured, low-context asynchronous operational playbooks that minimized cognitive friction for contributors.

Full-Stack ERPNext & Automation Engineer

Cape Town

FARO AFRICA

Aug 2024 – Nov 2025

- Refactored and extended ERPNext via Python and JavaScript server hooks, optimizing pricing matrices and BI reporting.
- Architected migration of internal Retool workflows to Expo/React Native, reducing vendor licensing overhead.
- Built custom NFC scanner utilities within Expo for instantaneous physical inventory syncs.
- Provisioned AWS infrastructure with Pulumi, deploying Inngest event-driven functions and PayloadCMS.
- Diagnosed and resolved issues in a large existing ERPNext installation; introduced LLM-assisted debugging workflows.

Enterprise Infrastructure & Tooling Engineer

Remote

INGENICS DIGITAL GMBH (THROUGH VIVA OUTSOURCING)

Mar 2023 – May 2024

- Built modular Python utility structures and automation scripts for internal configuration schema validation.
- Designed data handling pipelines ensuring data transfer consistency across runtime platforms using declarative specifications.
- Developed Python hardware mocks enabling faster development cycles with fewer hardware dependencies.

Junior Lecturer

Potchefstroom

NORTH-WEST UNIVERSITY

Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students.

Skills

Primary

Python, ERPNext/Frappe Framework, Expo (React Native), Inngest

Secondary

AWS, Pulumi (IaC), PostgreSQL (RLS), Docker

Foundational

SQL, Object-Oriented Design, Spec-Driven Development

Other

C++, FreeRTOS (Low-level integration support)

Education

B.Eng. Computer and Electronic Engineering

Potchefstroom

NORTH-WEST UNIVERSITY

2020

Focus on embedded systems, software engineering, and electronic design · Developed data analysis pipeline with Python/Pandas (linear regression, ML); built NFC payment emulator (Kotlin/Android)